LESSON 2.1

# Understand Statistical Questions

**Set B · Version A** · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can tell the difference between a statistical question and a non-statistical question.

**WHY THIS MATTERS** You've just been hired as the data analyst for a youth basketball league. The head coach hands you a list of questions the staff wants to investigate about their players. Some questions will give you interesting data with variety, while others have just one fixed answer. Your job: figure out which questions are worth investigating with data. Two of the coach's questions are: "How tall is our point guard?" and "How tall is each player on the roster?"

REMEMBER

## Understand Statistical Questions

**Statistical Questions & Variability** A statistical question is one you answer by collecting data, and the answers are different for different people or cases. A non-statistical question has just one fixed answer.

1. Statistical Question: Expects answers that vary across people, trials, or items.
2. Non-Statistical Question: Has one single fixed answer.
3. Data Collection: Gather varied data points to create a distribution

**Words:** **Statistical Question *(Pregunta estadística)*** **Non-statistical Question *(Pregunta no estadística)*** **Data *(Datos)*** **Variability *(Variabilidad)*** **Survey *(Encuesta)***

**1** **Circle the letter of the best answer. Show how you know.**

Which of these would collect data that VARIES (making it a statistical question)?

(A) How many siblings does each student in the class have?

(B) How many days are in one week?

(C) How many minutes are in one hour?

(D) What is 12 + 12?

**HINT** The capital word VARIES is the clue — you need the question whose answers change from student to student.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

A box has a base area of 10 square inches and a height of 4 inches. What is its volume?

(A) 2.5 in³
(B) 14 in³
(C) 40 in³
(D) 40 in²

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

**3** **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 6 cm, w = 2 cm and h = 3 cm?

(A) 11 cm³
(B) 18 cm³
(C) 36 cm³
(D) 36 cm²

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

**4** **Circle the letter of the best answer. Show how you know.**

A prism has a base area of 10 square feet and a volume of 40 cubic feet. What is its height?

(A) 4 feet
(B) 30 feet
(C) 50 feet
(D) 400 feet

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

**5** **Circle the letter of the best answer. Show how you know.**

A box measures 1/2 ft by 3 ft by 4 ft. What is its volume?

(A) 2 cubic feet
(B) 6 cubic feet
(C) 7.5 cubic feet
(D) 12 cubic feet

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

**6** **Circle the letter of the best answer. Show how you know.**

The school newspaper wants to write a story about the girls' soccer team. A reporter suggests asking: "How many goals has the team scored total?" The editor says they should ask: "How many goals did each player score this season?" Which question is statistical?

(A) 'How many goals has the team scored total?'

(B) Both are statistical

(C) 'How many goals did each player score this season?'

(D) Neither is statistical

SHOW YOUR WORK

EXPLAIN YOUR THINKING

**What pattern do you notice about statistical questions compared to non-statistical questions?**

SENTENCE STARTER Statistical questions expect ___ in the answers because ___. Non-statistical questions have only ___ answer because ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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## PART 1 Understand the idea

models and meaning

**1** **Write the name of the correct group on each line.**

Drag each scouting question into the right bin. A statistical question expects answers that VARY from player to player; a non-statistical question has one fixed answer.

**Groups:** Statistical | Not Statistical

_____ How many push-ups can each player on the team do in one minute?

_____ How many points does each starter average per game?

_____ How many seconds does each player take to sprint the court?

_____ How many free throws did each teammate make in practice?

_____ How many players are allowed on the court at once?

_____ What is the diameter of an official basketball hoop?

_____ How many quarters are played in one game?

**2** **Write the letter of the matching item on each line.**

Match each weak (non-statistical) scouting question to a stronger version that becomes statistical by adding variability.

_____ How many players are on the roster?

_____ What time does the game start?

_____ How long is the basketball court?

_____ How many points did the team score in total?

(A) How many minutes does each player on the roster play per game?

(B) What time does each fan arrive at the arena?

(C) How far from the hoop does each player shoot most often?

(D) How many points did each player score in the game?

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

The school newspaper wants to write a story about the girls' soccer team. A reporter suggests asking: "How many goals has the team scored total?" The editor says they should ask: "How many goals did each player score this season?" Why is the team total NOT a statistical question?

(A) Because totals are never useful

(B) Because it is about soccer

(C) Because you cannot count goals

(D) Because it has exactly one answer, with no variability

SHOW YOUR WORK

**4** **Circle the letter of the best answer. Show how you know.**

The school newspaper wants to write a story about the girls' soccer team. A reporter suggests asking: "How many goals has the team scored total?" The editor says they should ask: "How many goals did each player score this season?" Which of these is a statistical question?

(A) How tall am I?

(B) What is today's date?

(C) How many minutes are in an hour?

(D) How tall are the students in my class?

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

The team analyst writes four questions for the scouting report. Which one is a statistical question?

(A) How many points is a single free throw worth?

(B) What is the height of the basketball rim?

(C) How many rebounds does each player grab per game?

(D) How many timeouts does each team get per half?

SHOW YOUR WORK

**6** **Circle the letter of the best answer. Show how you know.**

Why does the question "How many minutes did each athlete warm up before the game?" count as a statistical question?

(A) Because the answer is measured in minutes

(B) Because the warm-up times will vary from one athlete to another

(C) Because every athlete warmed up for exactly the same time

(D) Because it is about a sports team

SHOW YOUR WORK

EXPLAIN YOUR THINKING

**What pattern do you notice about statistical questions compared to non-statistical questions?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help